



TEAM ASTROPHAGE

TECHATHON 2.0 · NAARI SHAKTI

Quantifying Ion Energy Flux Across Plasma Sheaths

A non-intrusive optical diagnostic framework that fuses time-resolved measurements with first-principles sheath modeling to reconstruct ion energy flux at plasma–surface boundaries.

Nithisha V · Ajayaditya L · Denistan B · Danushika N

Plasma Technologies for Industry & Society · AIC RRCAT π -Hub Foundation, DAE



THE PROBLEM · WHY IT MATTERS

Ion energy flux decides how plasma hardware heats, erodes and lasts

At every plasma–surface boundary, ions accelerate across a thin charged layer — the sheath — and strike the wall. The energy they deposit drives three outcomes that make or break plasma systems:



Surface heating

Localized thermal loads set cooling limits and cap the power a device can sustain.



Erosion & lifetime

Ion bombardment sputters walls and electrodes — the dominant wear mechanism.



Efficiency

Energy lost to surfaces is energy not delivering thrust, etch precision or confinement.

WHERE IT BITES



Electric propulsion thrusters



Plasma etch reactors



Fusion plasma-facing walls



PROBLEM STATEMENT · THE GAP

We can predict ion energy flux — but we cannot reliably measure it

First-principles models (fluid and kinetic) estimate flux from sheath potential and plasma parameters. **But those predictions stay weakly validated under realistic, non-equilibrium and time-varying conditions** — because the experimental tools fall short:



Intrusive probes

- ✗ Physically inserted into the plasma — they perturb the very sheath being measured
- ✗ Distort potential, density and the local field, biasing every derived quantity
- ✗ Cannot survive or resolve harsh, transient, high-flux environments



Non-invasive optics, today

- Capture only partial observables — one technique, one slice of the picture
- Insufficient on their own to reconstruct energy deposition at the surface
- Rarely time-resolved, so transient dynamics are averaged away



The result: no widely accepted, experimentally validated framework for quantifying ion energy transfer at plasma–surface boundaries.



PROPOSED SOLUTION

Infer energy flux by fusing non-perturbing optics with sheath physics

We treat ion energy flux as an inverse problem: measure what optics can see without disturbing the plasma, constrain it with first-principles physics, and reconstruct the quantity we actually need.



01

MEASURE

Capture non-perturbing optical observables — electron temperature and density, and ion velocity distributions.



02

MODEL

Constrain the problem with sheath theory and kinetic simulations as physical priors.



03

RECONSTRUCT

Solve the inverse problem for ion energy flux — resolved in space and in time.



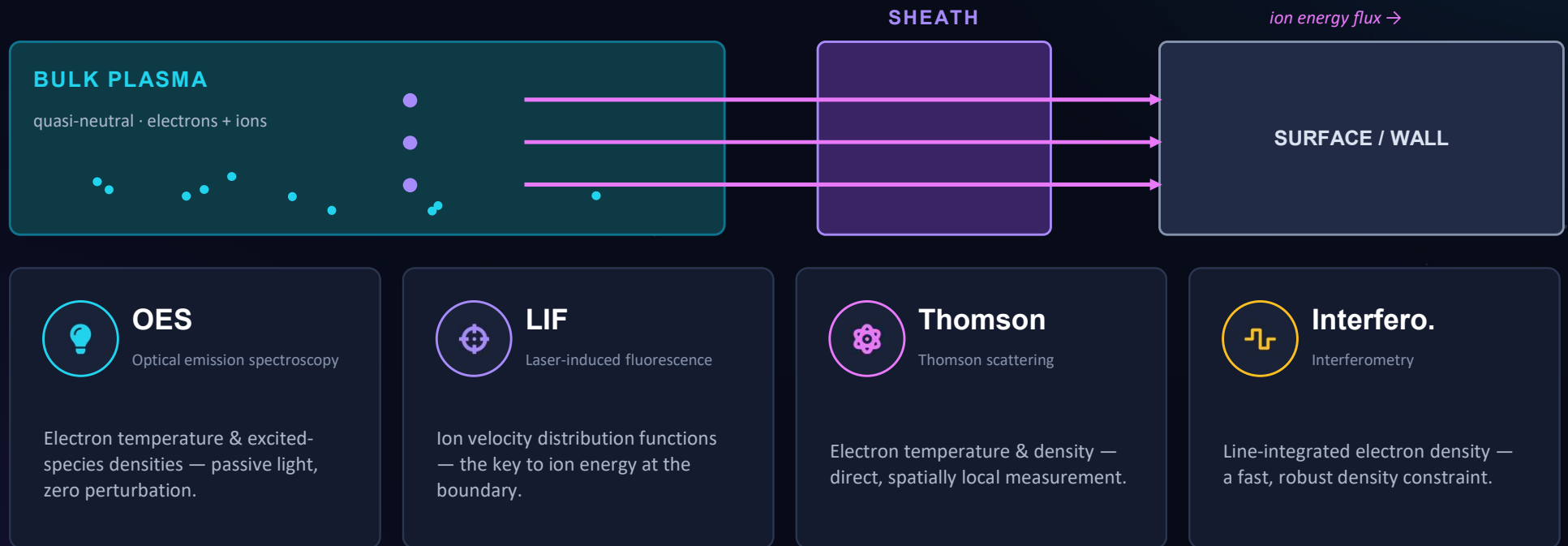
Outcome: validated, spatially and temporally resolved energy-deposition estimates at the plasma–surface boundary — without a single probe in the plasma.



PROPOSED SOLUTION · THE TOOLKIT

Four complementary non-perturbing diagnostics, one physical picture

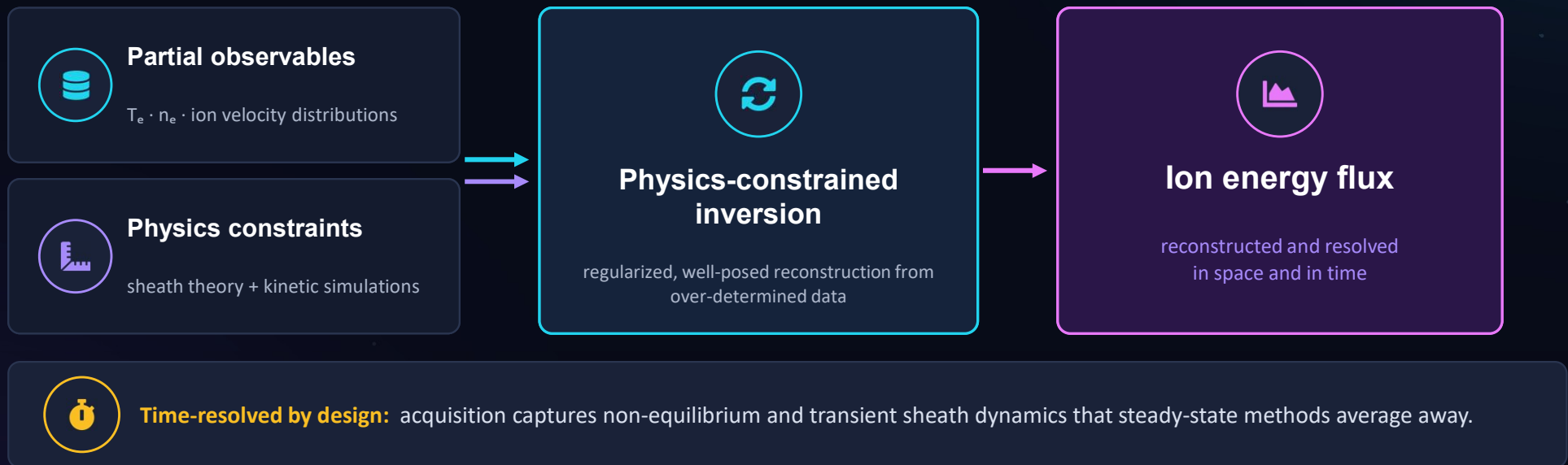
Each technique is individually mature; together they over-determine the plasma state — the redundancy that makes the reconstruction well-posed.





A physics-constrained inverse framework recovers the unmeasurable

Ion energy flux is not directly observable. We pose it as a regularized inverse problem — letting physics fill the gap that any single measurement leaves open.





INNOVATION & UNIQUENESS

Replacing the probe with a physics-constrained sensor fusion engine

The central innovation: instead of one perturbing measurement, we fuse multiple non-perturbing diagnostics and regularize with physics to recover a quantity none of them sees alone.



Conventional intrusive probes

- ✗ Perturbs and distorts the sheath it measures
- ✗ Single, limited observable
- ✗ Steady-state / time-averaged only
- ✗ Cannot yield ion energy flux directly
- ✗ Biases lifetime and erosion predictions



Astrophage framework

- ✓ Fully non-perturbing — optics only
- ✓ Fuses four complementary diagnostics
- ✓ Time-resolved — captures transients
- ✓ Reconstructs energy flux via physics-constrained inversion
- ✓ Validated, defensible predictions



TECHNICAL APPROACH & FEASIBILITY

A staged path that de-risks before it scales

We begin where confidence is highest — validated models — then add experiment, then resolution. Each stage produces a checkable result before the next.

1



Models & inversion

Analytical sheath models and kinetic simulations; build and validate the inverse framework on synthetic data.

2



Controlled lab trials

Stand up OES, LIF and interferometry on a controlled plasma source; test reconstruction against measurements.

3



High-resolution scale-up

Add Thomson scattering with advanced laser and vacuum infrastructure; validate under realistic conditions.



Why it is feasible

Every diagnostic is individually proven; the novelty is the fusion and inversion. Staged validation keeps each step checkable and low-risk.



Assumptions still being validated

Inversion well-posedness under measurement noise; optical access geometry on target hardware; cross-diagnostic calibration.



MARKET RELEVANCE & IMPLEMENTATION

A diagnostics layer for large, growing plasma-driven markets

We are not selling thrusters or etchers — we are the measurement and qualification layer they depend on. A high-value slice of three expanding markets:



Electric & space propulsion

~\$20B

space propulsion by 2030 (~12% CAGR)

70,000+ satellites projected by 2030, ~80% using electric propulsion that erodes under ion flux.



Semiconductor plasma etch

\$37B+

etch equipment by 2030

Plasma / dry etch leads the market; sub-5 nm yield hinges on atomic-scale process control.



Fusion & advanced reactors

Emerging

high-value research market

Plasma-facing erosion and heat load remain a top unsolved barrier to viable fusion.

Our wedge is the diagnostic & qualification layer serving all three. · Market figures: third-party industry forecasts, 2025.



COMMERCIALIZATION & INCUBATION

From validated framework to a deployable diagnostic product



Product

A non-intrusive diagnostic package: optical hardware bundled with the physics-constrained inversion software.



Customers

Thruster and EP developers (qualification), semiconductor fabs (process control), and fusion / research labs.



Revenue model

Diagnostic systems, inversion-software licensing, and qualification-as-a-service for hardware developers.

INCUBATION PATHWAY



Incubate at AIC π -Hub



Access RRCAT laser & vacuum infra



Partner pilot & qualification



Productize & commercialize



ROADMAP & TIMELINE

A phased path to a validated proof-of-concept

The computational and experimental tracks run in parallel, converging on a validated proof-of-concept on a controlled plasma source.

Months 0–3

Foundation & Modeling

Sheath + kinetic models; formulate the inversion; validate on synthetic data.



Months 6–9

Integration & First Results

Run the inversion on real measurements — first ion-energy-flux reconstructions.



Months 3–6

Diagnostic Bring-Up

OES + interferometry on a controlled source; begin LIF integration.



Months 9–12

Proof-of-Concept Validation

Benchmark accuracy, quantify uncertainty, demonstrate time-resolved capability.



SCALE-UP Thomson scattering + advanced laser/vacuum, high-resolution validation under realistic conditions, then partner pilot & productization.



TEAM CAPABILITY

Team Astrophage

A four-person team spanning the full stack this problem demands — from optics and plasma physics to experiment and computation.



Nithisha V

Optical Diagnostics & Spectroscopy

OES / LIF design, optical setup and signal interpretation.



Ajayaditya L

Plasma Modeling & Kinetic Simulation

Sheath theory, kinetic simulations and the inversion physics.



Denistan B

Experiments, Vacuum & Lasers

Plasma source, vacuum systems and laser-diagnostic integration.



Danushika N

Inverse Methods, Data & Software

Reconstruction algorithms, calibration and the data pipeline.



TEAM ASTROPHAGE

THE VISION

An in-situ standard for measuring sheath energy transfer — turning probe-free optics into trustworthy lifetime, erosion and thermal predictions across every plasma technology.

WHAT WE ARE SEEKING



Incubation & mentorship

DIA ecosystem support and scientific mentorship to mature the framework.



Infrastructure access

Advanced laser and vacuum facilities for high-resolution validation.



A pilot partner

An electric-propulsion or fab partner to qualify the diagnostic in the field.



Thank You

Team Astrophage · Quantifying Ion Energy Flux Across Plasma Sheaths

With gratitude to the AIC RRCAT π -Hub Foundation, the Department of Atomic Energy & RRCAT, the DAE Incubator Alliance, the Atal Innovation Mission (NITI Aayog), and our mentors and families.

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